CELLULAR PARABIOSIS & ORGAN SPECIFICITY OF METASTASES: TESTING A HYPOTHESIS

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Premises:

The aim of this project is to test the hypothesis that cellular parabiosis (i.e., the trans-cellular functional complementation) determines tissue selectivity of metastases (M. Radman, Open Biol. 2019).

This hypothesis posits that the latency period for carcinomas in solid tissues (average about 20 yr, e.g., after atomic bomb explosions) is due to phenotypic suppression of malignant cells by healthy neighboring cells. Each potential metastatic cell homing among healthy cells of the same organ has to pass the same kind of 20 yr latency as the original cell of the primary tumour. Tumour suppressing functions provided by cells of the primary organ encountered in other organs will also suppress metastatic growth of invading cells; only organs unable to provide phenotypic suppression allow can home metastases. This might be the reason for the rarity of metastases in the organ of the primary tumour. Parabiotic suppression of malignancy requires the structures (e.g., TnTs and GJ) and specific contents (e.g., expression of specific tumour suppressors (TS) lost in primary tumour). Hence, the tissue selectivity of metastases.

In addition to this tumor-suppressing c-parabiosis, there is also a tumor-stimulating c-parabiosis (e.g., the case of cancer-associated fibroblasts (CAF) in the stroma) based on phenotypic suppression of diverse metabolic deficiencies in tumor cells. These metabolic defects are selectively purified by selective cell death (anoikis) of tumor cells while travelling through the bloodstream. Thus, only when tissues provide for specific tumor suppressive functions lost in tumors will c-parabiosis suppress metastatic growth; otherwise c-parabiosis may increase the metabolic fitness of emerging metastases (e.g., the effect of CAFs).

The prediction that organs expressing many TS will rarely home diverse metastases (and those expressing few TS, e.g., the liver, will be frequent home for metastases) seems to be substantiated (Tinka's work). Now, we want to use DNA and RNA sequencing to identify TS functions lost in primary tumour and provided by tissues resilient to metastases.

Prepare the materials and collect relevant informations:

(1) Get fresh biopsies of a primary tumour and its one, or more, metastases.
(2) Get also biopsies of healthy tissue of primary and metastatic tumours including the stromal fibroblasts next to the primary tumour.

(3) DNA sequence (directly, no culturing) several sections of the primary tumour (single cell sequencing would be ideal).
(4) Get the transcriptomes (and proteomes?) from healthy parts sites/tissues of original primary and secondary (home of metastases)
(5) DNA sequence (directly, no culturing) the metastasis (one or more).
(6) from (2), identify driver mutations (in tumour suppressors and oncogenes)
(7) Get the transcriptomes (and proteomes?) from healthy parts (sites/tissues) of the original primary and secondary (home of metastases) malignancies.

(B) The Stocker experiment:

(8) Make limited primary tumour cultures from (1).
(9) Test individual cells from primary and metastatic tumors for growth on diverse human fibroblasts (George Klein's work): identify suppressing and non-suppressing fibroblast.
(10) Sequence DNA (and DNA methylation?) and get transcriptomes and proteomes of these suppressing and non-suppressing fibroblasts. Check whether the expression of genes in suppressing fibroblasts relates to tumour genes bearing driver mutations.

(C) Reconstruction experiment:

If the expression in fibroblasts of the TS gene(s) corresponds to driver genes lost in tumour (primary and seconday) by mutation, then:
(11) Knock-out such gene(s) in suppressing fibroblasts & knock-in the same genes with active promotor in non-suppressing fibroblasts, then:
(12) Test genetically modified fibroblasts in Stocker expt (see 9), i.e., whether the capacity for suppression of tumour growth is inversed by the two genetic modifications (from 11).

A SHORTER VERSION: MAKE USE OF GEORGE KLEIN'S WORK

(13) G. Klein used Stocker expt with prostate cancer PC3 cells with nuclei labelled by H2A-mRFP histone with suppressing and non-suppressing fibroblasts with analyzed transcriptomes. We could knock-out and knock-in candidate genes to test for changed malignant suppressivity.

IS "FEEDER LAYER" TECHNIQUE A CASE OF C-PARABIOSIS?

(14) Feeder layer - usually fibroblasts irradiated with low dose of X-rays - is a classical technique to grow some "difficult" cells, e.g., embryonic stem cells. Direct contact is required for the effect. Low X-ray irradiation is among the TnT-stimulating stresses. Hence, the provision by the feeder

layer of factors missing in standard media (for particular cells) could be a case of c-parabiosis (like CAFs for tumours).

DATA TO LOOK FOR IN (B) AND (C):

(i) Heterogeneity of DNA sequence within the primary tumour - compare with heterogeneity of sequence in the metastasis to see the consequence of selection for fitness during the search for homing.

(ii) Identification of driver mutation (genes) of the primary malignancy and expression of functional genes suppressing the malignancy.

(iii) Identify in Stocker experiment the means (TnT, GJ, exosomes, etc) by which phenotypic suppression occurs.

THINK HOW TO USE THIS KNOWLEDGE TO *HEAL CANCER BY PHENOTYPIC REVERSION OF PRIMARY AND METASTATIC MALIGNANCY BACK TO PHENOTYPIC NORMALITY (IN ADDITION TO "DIVIDE & CONQUER" STRATEGY)!*